AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Chemistry Paper I

Time allowed: 35 minutes Marks 25

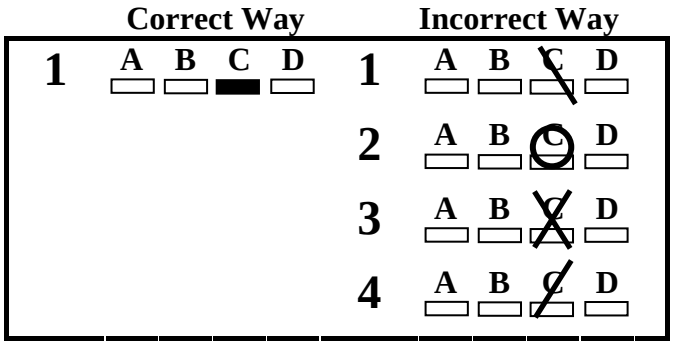
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

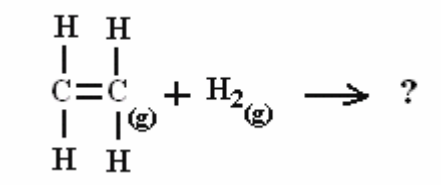
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. Which of the following is used as moderator in a nuclear reactor?

   A. Soft water
   B. Hard water
   C. Heavy water
   D. Mineral water

2. Which of the following products is formed in the given reaction?



   A. Ethane
   B. Ethene
   C. Ethyne
   D. Ethanol

3. How many lone pairs are present in a water molecule?

   A. 2
   B. 3
   C. 4
   D. 6

4. Which type of reaction occurs in the following reaction?

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$$

   A. Exothermic
   B. Endothermic
   C. Neutralization
   D. Decomposition

5. Silica gel enclosed in a small satchel is used in medicine bottles due to its property of

   A. moisturisation.
   B. preservation.
   C. absorption of heat.
   D. absorption of water vapours.

6. Which of the following are the possible products of the reaction of silicon dioxide with concentrated potassium hydroxide?

   A. Potassium silicate and water
   B. Potassium silicate and hydrogen
   C. Potassium hydroxide and silicon
   D. Potassium hydroxide and hydrogen

7. Which of the following reactions results in the formation of urea?

A. Ammonia + Water
B. Ammonia + Carbonate
C. Ammonia + Nitric acid
D. Ammonia + Carbon dioxide

8. Which of the following is an oxidation reaction?

A. $C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$
B. $ZnO_{(s)} + H_{2(g)} \rightarrow Zn_{(s)} + H_2O_{(l)}$
C. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$
D. $KMnO_{4(aq)} + 2H_2O_{(l)} + 5SO_{2(g)} \rightarrow K_2SO_{4(aq)} + 2MnSO_{4(aq)} + 2H_2SO_{4(aq)}$

9. A suspension of MgO is used to control the acidity of the stomach because it is

A. a basic oxide.
B. a neutral oxide.
C. an acidic oxide.
D. an amphoteric oxide.

10. The chemical reaction given below shows that sulphuric acid is a

$$H_2SO_{4(aq)} \xrightarrow{\text{Water}} 2H^+_{(aq)} + SO^{-2}_{4(aq)}$$

A. monobasic acid.
B. dibasic acid.
C. tribasic acid.
D. tetrabasic acid.

11. Which of the following substances brings tears in the eyes while slicing an onion?

A. Carbon
B. Sulphur
C. Oxygen
D. Nitrogen

12. The following warning sign is always displayed on the containers containing sulphuric acid.



It indicates that sulphuric acid is

A. oily.
B. corrosive.
C. carcinogenic.
D. hygroscopic.

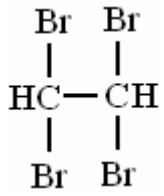
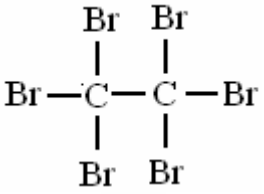
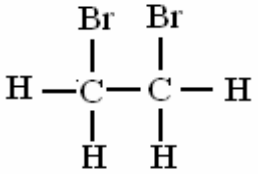
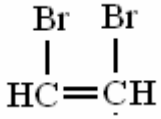
13. Which of the following two ions are formed when hydrogen chloride gas dissolves in water?

A. $H_3O^+ + Cl^-$
B. $HOH + Cl^-$
C. $H^+ + Cl^-$
D. $H^+ + OH^-$

14. Which of the following products is formed when silver nitrate and sodium bromide act as reactants?

A. $AgBr + NaNO_3$
B. $AgNO_3 + NaBr$
C. $AgNO_2 + NaBr + H_2$
D. $AgBr_2 + NaNO_2 + H_2O$

15. Which of the following compounds will be formed when ethyne reacts with bromine in the presence of metallic halide as a catalyst?

 A	 B	 C	 D

16. Which of the following compounds contains a double bond?

A. Butane
B. Butene
C. Propyne
D. Propane

17. Which of the following is identified as the first step in the refining process of crude oil?

A. Boiling
B. Heating
C. Evaporation
D. Neutralization

18. Which of the following is used in stimulating the growth of potatoes?

A. Ethane
B. Ethene
C. Ethyne
D. Ethanol

19. Meat and fish are preserved by the addition of

A. sodium chloride.
B. sodium sulphite.
C. sulphur dioxide.
D. sulphurous acid.

20. Which of the following is commonly called“metha soda”(sweet soda)?

A. NaCl
B. NaOH
C. Na_2CO_3
D. $NaHCO_3$

21. In the process of extraction of copper, blistered copper acts as

A. a cathode.
B. an anode.
C. an electrolyte.
D. an electrolytic cell.

22. Which of the following is used to decrease the melting point of bauxite in the extraction of aluminium during Hall’s process?

A. Carbon
B. Fluorospar
C. Molten alumina
D. Molten cryolite

23. Which of the following is also known as “stainless steel”?

A. Nickel steel
B. Silicon steel
C. Chromium steel
D. Manganese steel

24. When carbon contents are removed in the form of CO and CO_2 from a blast furnance, the left over is

A. pig iron.
B. cast iron.
C. wrought iron.
D. alloy steel.

25. When non-metals react with oxygen they form

A. acidic oxide.
B. basic oxide.
C. neutral oxide.
D. amphoteric oxide.

END OF PAPER

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